

Predictive Analytics Using Historical Data

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Abstract

This project analyzes historical datasets to identify trends and forecast future outcomes using Power BI and predictive analytics techniques. It includes data collection, cleaning, exploratory analysis, forecasting, dashboard development, and evaluation.

Introduction

The project demonstrates how historical business data can be transformed into actionable insights using Power BI dashboards and predictive models.

Objectives

- Analyze historical data.
- Clean and preprocess datasets.
- Build predictive models.
- Create an interactive Power BI dashboard.
- Forecast future trends.

Methodology

Data Collection → Data Cleaning → Exploratory Data Analysis → Predictive Modeling → Dashboard Development → Forecasting & Evaluation.

Dashboard KPIs

Total Sales, Total Profit, Total Orders, Average Sales, Profit Margin, Forecasted Sales, Model Accuracy, MAE, and RMSE.

Conclusion

The project demonstrates practical business intelligence and predictive analytics skills, enabling data-driven decision-making through interactive dashboards.